

PRACTICE QUESTIONS ON

# CAPITAL MARKET THEORY

*Numerical Problems with Solutions*

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A Practice-Oriented Numerical Companion

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**Securities Analysis and Portfolio Management (SAPM)**

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## About This Question Bank

This question bank compiles the numerical problems from the accompanying teaching notes on Capital Market Theory into a single, practice-oriented resource. Each problem is presented in a consistent four-part format: a Learning Objective, the Problem statement, a step-by-step Solution, and the Final Answer / Interpretation. Original numerical data, formulas, and calculation logic have been preserved throughout.

The material is organized into eight sections, moving from single-asset return and risk measurement through portfolio theory, the Capital Market Line, the Security Market Line / CAPM, and finally risk-adjusted performance evaluation:

- Section 1: Return and Risk Measurement
- Section 2: Holding Period Return and Effective Annual Return
- Section 3: Portfolio Risk, Return, and Correlation
- Section 4: Covariance, Correlation, Beta, Alpha, and Risk Decomposition
- Section 5: Efficient Frontier Problems
- Section 6: Capital Market Line Problems
- Section 7: Security Market Line and CAPM Valuation
- Section 8: Risk-Adjusted Performance Evaluation

*Students are encouraged to attempt each problem independently before referring to the solution, and to focus on the reasoning in the Final Answer / Interpretation section, which links the arithmetic back to the underlying investment decision.*

## Section 1: Return and Risk Measurement

### Problem 1.1: Single-Period Return

#### Learning Objective:

To calculate the single-period return of an investment from a series of year-end prices and interpret the variability of year-to-year investment performance.

#### Problem:

The year-end prices of an investment over five years are given below. Calculate the single-period return for each year.

| Time | Price |
|------|-------|
| 0    | 100   |
| 1    | 110   |
| 2    | 108   |
| 3    | 130   |
| 4    | 154   |
| 5    | 145   |

Formula:

$$R_t = (P_t - P_{t-1}) / P_{t-1}$$

#### Solution:

Year 1

$$R_1 = (110 - 100) / 100 = 10/100 = 10.00\%$$

Year 2

$$R_2 = (108 - 110) / 110 = -2/110 = -1.82\%$$

Year 3

$$R_3 = (130 - 108) / 108 = 22/108 = 20.37\%$$

Year 4

$$R_4 = (154 - 130) / 130 = 24/130 = 18.46\%$$

Year 5

$$R_5 = (145 - 154) / 154 = -9/154 = -5.84\%$$

**Final Answer / Interpretation:**

| Year | Return |
|------|--------|
| R1   | 10.00% |
| R2   | -1.82% |
| R3   | 20.37% |
| R4   | 18.46% |
| R5   | -5.84% |

The investment generated positive returns in three years (Years 1, 3, and 4) and negative returns in two years (Years 2 and 5). Return performance is uneven and must be evaluated along with risk.

## Problem 1.2: Arithmetic Mean Return

### Learning Objective:

*To calculate the arithmetic mean return as the simple average of periodic returns, used to estimate the expected return for a single future period.*

### Problem:

Using the annual returns computed in Problem 1.1 (10.00%, -1.82%, 20.37%, 18.46%, -5.84%), calculate the arithmetic mean return.

Formula:

$$\bar{R} = (\sum R_t) / n$$

### Solution:

$$\begin{aligned}\bar{R} &= (10.00 + (-1.82) + 20.37 + 18.46 + (-5.84)) / 5 \\ \bar{R} &= 41.17 / 5 = 8.23\%\end{aligned}$$

### Final Answer / Interpretation:

**$\bar{R} = 8.23\%$**

The investment earned an average annual return of 8.23%. However, this is only a simple average and does not show the actual compounded growth of wealth.

## Problem 1.3: Geometric Mean Return

### Learning Objective:

*To calculate the geometric mean return (compound annual growth rate) and show the actual rate at which investor wealth grows over time, in contrast with the arithmetic mean.*

**Problem:**

Using the initial price  $P_0 = 100$  and the final price  $P_5 = 145$  over  $n = 5$  years, calculate the geometric mean return.

Formula:

$$R_G = (P_n / P_0)^{1/n} - 1$$

**Solution:**

$$\begin{aligned} P_0 &= 100, P_5 = 145, n = 5 \\ R_G &= (145/100)^{1/5} - 1 = (1.45)^{0.20} - 1 \\ R_G &= 1.0771 - 1 = 0.0771 = 7.71\% \end{aligned}$$

**Final Answer / Interpretation:**

**RG = 7.71%**

The investor's wealth compounded at 7.71% per year. This is lower than the arithmetic mean because fluctuations in returns reduce compounded growth.

**Problem 1.4: Standard Deviation of Returns**

**Learning Objective:**

*The standard deviation of returns is commonly used as a measure of dispersion around the average return and is a measure of total investment risk.*

**Problem:**

Using the annual returns from Problem 1.1 and the arithmetic mean return of 8.23%, calculate the standard deviation of returns.

Formula (sample standard deviation):

$$\sigma = \sqrt{[\sum (R_t - \bar{R})^2 / (n - 1)]}$$

**Solution:**

| Return | Deviation from Mean | Squared Deviation |
|--------|---------------------|-------------------|
| 10.00% | 1.77%               | 3.13              |

| Return | Deviation from Mean | Squared Deviation |
|--------|---------------------|-------------------|
| -1.82% | -10.05%             | 101.00            |
| 20.37% | 12.14%              | 147.38            |
| 18.46% | 10.23%              | 104.65            |
| -5.84% | -14.07%             | 197.96            |

$$\Sigma(R_t - \bar{R})^2 = 554.12$$

$$\sigma = \sqrt{554.12 / 4} = \sqrt{138.53} = 11.77\%$$

**Final Answer / Interpretation:**

**$\sigma = 11.77\%$**

The annual returns fluctuate by approximately 11.77 percentage points around the average return of 8.23%. A higher standard deviation indicates a higher risk.



## Section 2: Holding Period Return and Effective Annual Return

### Problem 2.1: Six-Month Investment: HPR and EAR

#### Learning Objective:

*To calculate the holding period return for a six-month investment and convert it to an effective annual return and understand why annualization is necessary for meaningful comparison.*

#### Problem:

An investor purchases a security for ₹97.36 that redeems at ₹100 after six months ( $n = 0.5$  years). Calculate (a) the Holding Period Return (HPR) and (b) the Effective Annual Return (EAR).

Formulas:

$$HPR = (FV - PV) / PV = FV/PV - 1$$
$$EAR = (1 + HPR)^{1/n} - 1$$

#### Solution:

##### (a) Holding Period Return

$$HPR = 100/97.36 - 1 = 1.0271 - 1 = 2.71\%$$

##### (b) Effective Annual Return

$$EAR = (1 + 0.0271)^{1/0.5} - 1 = 1.05497 - 1 = 5.497\%$$

#### Final Answer / Interpretation:

**HPR = 2.71%, EAR = 5.497%**

An investor purchasing the security for ₹97.36 and receiving ₹100 after six months earns a 2.71% return during the six-month holding period. This is equivalent to an annual return of 5.497%.

### Problem 2.2: One-Year Investment: HPR and EAR

#### Learning Objective:

*To calculate the holding period return and effective annual return for a one-year investment and observe that the two measures coincide when the holding period equals one year.*

#### Problem:

An investor purchases a security for ₹95.52 that redeems at ₹100 after one year ( $n = 1$  year). Calculate the HPR and the EAR.

$$HPR = FV/PV - 1$$

$$EAR = (1 + HPR)^{1/n} - 1$$

**Solution:**

$$HPR = 100/95.52 - 1 = 1.0469 - 1 = 4.69\%$$

$$EAR = (1 + 0.0469)^1 - 1 = 4.69\%$$

**Final Answer / Interpretation:**

**HPR = EAR = 4.69%**

Since the investment horizon is already one year, HPR and EAR are identical.

**Problem 2.3: Twenty-Five-Year Investment: HPR and EAR**

**Learning Objective:**

*To calculate the holding period return and effective annual return for a long-horizon (25-year) investment and demonstrate why a large cumulative return can correspond to a modest annualized return.*

**Problem:**

An investor purchases a security for ₹23.30 that redeems at ₹100 after twenty-five years ( $n = 25$  years). Calculate the HPR and the EAR.

$$HPR = FV/PV - 1$$

$$EAR = (1 + HPR)^{1/n} - 1$$

**Solution:**

$$HPR = 100/23.30 - 1 = 4.2918 - 1 = 3.2918 = 329.18\%$$

$$EAR = (1 + 3.2918)^{1/25} - 1 = 1.0600 - 1 = 6.00\%$$

**Final Answer / Interpretation:**

**HPR = 329.18%, EAR = 6.00%**

Although the investment generated a cumulative return of 329.18% over the 25-year holding period, the annualized return is only 6%.

| Maturity | HPR   | EAR    |
|----------|-------|--------|
| 0.5 Year | 2.71% | 5.497% |

|          |         |       |
|----------|---------|-------|
| 1 Year   | 4.69%   | 4.69% |
| 25 Years | 329.18% | 6.00% |

At first glance, the 25-year investment appears dramatically superior on an HPR basis, but this conclusion is misleading because the returns are measured over different investment horizons. Once converted to Effective Annual Returns, the difference across the three investments is far smaller than it first appears, which is why returns measured over different horizons cannot be directly compared without annualizing.

## Section 3: Portfolio Risk, Return, and Correlation

### Problem 3.1: Portfolio Return (Two-Asset Portfolio)

#### Learning Objective:

To calculate portfolio return as the weighted average of the expected returns of the assets included in the portfolio.

#### Problem:

Asset L has an expected return of 10%, and Asset H has an expected return of 20%. An investor allocates 50% to Asset L and 50% to Asset H. Calculate the portfolio return. (This weighting and these returns are held constant in Problems 3.2 to 3.4, where only the correlation between the two assets changes.)

| Asset | Expected Return | Standard Deviation |
|-------|-----------------|--------------------|
| L     | 10%             | 16%                |
| H     | 20%             | 24%                |

$$R_p = w_L R_L + w_H R_H$$

#### Solution:

$$R_p = (0.5)(10\%) + (0.5)(20\%)$$

$$R_p = 5\% + 10\% = 15\%$$

#### Final Answer / Interpretation:

**R<sub>p</sub> = 15%**

Since portfolio weights and expected returns remain unchanged, portfolio return will remain 15% in all three correlation scenarios examined in Problems 3.2 to 3.4.

### Problem 3.2: Portfolio Risk – Perfect Positive Correlation ( $\rho = +1.00$ )

#### Learning Objective:

To calculate portfolio risk for a two-asset portfolio under perfect positive correlation and interpret the absence of diversification benefit.

#### Problem:

Using the data for Assets L and H above (weights 50%/50%), calculate portfolio risk when the correlation between the two assets is  $\rho = +1.00$ .

$$\sigma_p = \sqrt{w_L^2 \sigma_L^2 + w_H^2 \sigma_H^2 + 2w_L w_H \rho_{LH} \sigma_L \sigma_H}$$

**Solution:**

$$\begin{aligned}\sigma_p &= \sqrt{(0.5)^2(16)^2 + (0.5)^2(24)^2 + 2(0.5)(0.5)(1)(16)(24)} \\ \sigma_p &= \sqrt{64 + 144 + 192} = \sqrt{400} = 20\%\end{aligned}$$

**Final Answer / Interpretation:**

**$\sigma_p = 20\%$**

The two assets move perfectly together. There is no diversification benefit—the portfolio risk is simply the weighted average of the two asset risks.

### Problem 3.3: Portfolio Risk – Positive Imperfect Correlation ( $\rho = +0.65$ )

**Learning Objective:**

*To calculate portfolio risk for a two-asset portfolio under imperfect positive correlation and interpret the resulting diversification benefit.*

**Problem:**

Using the same data for Assets L and H (weights 50%/50%), calculate portfolio risk when the correlation between the two assets is  $\rho = +0.65$ .

$$\sigma_p = \sqrt{w_L^2 \sigma_L^2 + w_H^2 \sigma_H^2 + 2w_L w_H \rho_{LH} \sigma_L \sigma_H}$$

**Solution:**

$$\begin{aligned}\sigma_p &= \sqrt{(0.5)^2(16)^2 + (0.5)^2(24)^2 + 2(0.5)(0.5)(0.65)(16)(24)} \\ \sigma_p &= \sqrt{64 + 144 + 124.8} = \sqrt{332.8} = 18.24\%\end{aligned}$$

**Final Answer / Interpretation:**

**$\sigma_p = 18.24\%$**

The assets still tend to move in the same direction, but not perfectly. As a result, portfolio risk falls from 20% to 18.24%. This reduction is the diversification benefit.

### Problem 3.4: Portfolio Risk – Negative Correlation ( $\rho = -0.65$ )

**Learning Objective:**

*To calculate portfolio risk for a two-asset portfolio under negative correlation and demonstrate the strongest diversification benefit among the three scenarios.*

**Problem:**

Using the same data for Assets L and H (weights 50%/50%), calculate portfolio risk when the correlation between the two assets is  $\rho = -0.65$ .

$$\sigma_p = \sqrt{w_L^2 \sigma_L^2 + w_H^2 \sigma_H^2 + 2w_L w_H \rho_{LH} \sigma_L \sigma_H}$$

**Solution:**

$$\begin{aligned}\sigma_p &= \sqrt{(0.5)^2(16)^2 + (0.5)^2(24)^2 + 2(0.5)(0.5)(-0.65)(16)(24)} \\ \sigma_p &= \sqrt{64 + 144 - 124.8} = \sqrt{83.2} = 9.12\%\end{aligned}$$

**Final Answer / Interpretation:**

**$\sigma_p = 9.12\%$**

The assets tend to move in opposite directions. Losses in one asset are partially offset by gains in the other, so portfolio risk falls dramatically to 9.12% — the strongest diversification benefit among the three scenarios.

| Correlation ( $\rho$ ) | Portfolio Return | Portfolio Risk |
|------------------------|------------------|----------------|
| +1.00                  | 15.00%           | 20.00%         |
| +0.65                  | 15.00%           | 18.24%         |
| -0.65                  | 15.00%           | 9.12%          |

Portfolio return does not change across the three scenarios because portfolio weights and expected returns remain unchanged, but portfolio risk continuously declines as correlation decreases.

Diversification does not increase expected return — diversification reduces risk. This is the foundation of Markowitz Portfolio Theory and the Efficient Frontier.

## Section 4: Covariance, Correlation, Beta, Alpha, and Risk Decomposition

### Problem 4.1: Average Return of the Market and of Security A

#### Learning Objective:

To calculate the average returns of the market and a security over a historical period as a base measure for further risk-return analysis.

#### Problem:

The following seven years of market and Security A returns are given. Calculate the average return of the market and of Security A.

| Year | Market Return $R_m$ | Security A Return $R_A$ |
|------|---------------------|-------------------------|
| 1    | 12                  | 7                       |
| 2    | 13                  | 15                      |
| 3    | 18                  | 11                      |
| 4    | 16                  | 13                      |
| 5    | 21                  | 24                      |
| 6    | 11                  | 5                       |
| 7    | 14                  | 9                       |

$$\bar{R} = (\sum R_t) / n$$

#### Solution:

##### Market Average Return

$$\bar{R}_m = (12+13+18+16+21+11+14)/7 = 105/7 = 15\%$$

##### Security A Average Return

$$\bar{R}_A = (7+15+11+13+24+5+9)/7 = 84/7 = 12\%$$

#### Final Answer / Interpretation:

Market average return = 15%; Security A average return = 12%

### Problem 4.2: Variance and Standard Deviation of the Market and Security A

#### Learning Objective:

To calculate variance and standard deviation of both market and security returns as the base measures of total risk.

**Problem:**

Using the deviations of each year's return from its mean (shown below), calculate the variance and standard deviation of the market and of Security A.

| Year | R <sub>m</sub> | R <sub>A</sub> | R <sub>m</sub> - $\bar{R}_m$ | R <sub>A</sub> - $\bar{R}_A$ |
|------|----------------|----------------|------------------------------|------------------------------|
| 1    | 12             | 7              | -3                           | -5                           |
| 2    | 13             | 15             | -2                           | 3                            |
| 3    | 18             | 11             | 3                            | -1                           |
| 4    | 16             | 13             | 1                            | 1                            |
| 5    | 21             | 24             | 6                            | 12                           |
| 6    | 11             | 5              | -4                           | -7                           |
| 7    | 14             | 9              | -1                           | -3                           |

$$\sigma^2 = \Sigma(R_t - \bar{R})^2 / (n - 1)$$

**Solution:**

**Market Variance**

$$\Sigma(R_{mt} - \bar{R}_m)^2 = 9+4+9+1+36+16+1 = 76$$

$$\sigma_m^2 = 76/6 = 12.67$$

**Market Standard Deviation**

$$\sigma_m = \sqrt{12.67} = 3.56$$

**Security A Variance**

$$\Sigma(R_{At} - \bar{R}_A)^2 = 25+9+1+1+144+49+9 = 238$$

$$\sigma_A^2 = 238/6 = 39.67$$

**Security A Standard Deviation**

$$\sigma_A = \sqrt{39.67} = 6.30$$

**Final Answer / Interpretation:**

| Measure            | Market | Security A |
|--------------------|--------|------------|
| Variance           | 12.67  | 39.67      |
| Standard Deviation | 3.56%  | 6.30%      |

Security A's total risk (6.30%) is substantially higher than the market's volatility (3.56%).



### Problem 4.3: Covariance between Security A and the Market

#### Learning Objective:

To calculate covariance to determine whether a security's returns move together with, or opposite to, the market's returns.

#### Problem:

Using the deviation data from Problem 4.2, calculate the covariance between Security A's returns and the market's returns.

$$Cov_{A,m} = \Sigma(R_{At} - \bar{R}_A)(R_{mt} - \bar{R}_m) / (n - 1)$$

#### Solution:

$$\Sigma(R_{At} - \bar{R}_A)(R_{mt} - \bar{R}_m) = 15 + (-6) + (-3) + 1 + 72 + 28 + 3 = 110$$

$$Cov_{A,m} = 110/6 = 18.33$$

#### Final Answer / Interpretation:

**Cov(A,m) = 18.33**

Covariance is positive. Therefore, Security A generally moves in the same direction as the market.

### Problem 4.4: Correlation between Security A and the Market

#### Learning Objective:

The correlation coefficient is calculated to standardize covariance and measure the strength and direction of the relationship between a security's return and the market's return.

#### Problem:

Using  $Cov(A,m) = 18.33$ ,  $\sigma_A = 6.30$ , and  $\sigma_m = 3.56$ , calculate the correlation between Security A and the market.

$$\rho_{A,m} = Cov_{A,m} / (\sigma_A \sigma_m)$$

#### Solution:

$$\rho_{A,m} = 18.33 / (6.30 \times 3.56) = 18.33 / 22.43 = 0.82$$

**Final Answer / Interpretation:**

$$\rho(A,m) = 0.82$$

Security A has a strong positive relationship with the market. When the market moves up or down, Security A also tends to move in the same direction.

**Problem 4.5: Beta of Security A****Learning Objective:**

*To calculate beta, which measures the sensitivity of a security's return to the market's return and represents the security's systematic risk.*

**Problem:**

Using  $\text{Cov}(A,m) = 18.33$  and  $\sigma_m^2 = 12.67$ , calculate the beta of Security A.

$$\theta_A = \text{Cov}_{A,m} / \sigma_m^2$$

**Solution:**

$$\theta_A = 18.33 / 12.67 = 1.45$$

**Final Answer / Interpretation:**

$$\beta_A = 1.45$$

Security A is an aggressive security. If the market changes by 1%, Security A is expected to change by approximately 1.45% in the same direction.

**Problem 4.6: CAPM Expected (Required) Return of Security A****Learning Objective:**

*To calculate the CAPM-required return for a security based on the risk-free rate, market return, and beta.*

**Problem:**

The risk-free rate is 6.25%, and the market's average return is 15%. Security A's beta is 1.45. Calculate the CAPM expected return for Security A.

$$E(R_A) = R_f + \theta_A(\bar{R}_m - R_f)$$

**Solution:**

$$E(R_A) = 6.25 + 1.45(15 - 6.25)$$

$$E(R_A) = 6.25 + 1.45(8.75) = 6.25 + 12.66 = 18.91\%$$

**Final Answer / Interpretation:****E(RA) = 18.91%**

This is the return required to compensate investors for the level of systematic risk associated with Security A.

**Problem 4.8: Systematic Risk, Unsystematic Risk, and Percentage Risk Decomposition****Learning Objective:**

*To decompose a security's total risk into systematic (market-related) and unsystematic (firm-specific) components and express each as a percentage of total risk.*

**Problem:**

Security A's total variance is 39.67, its beta is 1.45, and the market variance is 12.67. Calculate the systematic risk, unsystematic risk, and the percentage composition of each.

$$\text{Systematic Risk} = \beta_A^2 \sigma_m^2$$

$$\text{Unsystematic Risk} = \sigma_A^2 - \text{Systematic Risk}$$

**Solution:****Systematic Risk**

$$= (1.45)^2(12.67) = 26.54$$

**Unsystematic Risk**

$$= 39.67 - 26.54 = 13.13$$

**Percentage Composition**

$$\text{Systematic \%} = 26.54/39.67 = 67\%; \text{ Unsystematic \%} = 13.13/39.67 = 33\%$$

**Final Answer / Interpretation:**

| Component         | Value | % of Total |
|-------------------|-------|------------|
| Systematic Risk   | 26.54 | 67%        |
| Unsystematic Risk | 13.13 | 33%        |

Security A has a beta of 1.45 (aggressive), but its actual average return is only 12%, while the CAPM required return is 18.91% (alpha = -6.91%). Since alpha is negative, Security A is not adequately compensating investors for its risk and therefore appears overpriced — not recommended for investment.

#### Problem 4.9: Risk Decomposition Expressed in Standard-Deviation Terms

##### Learning Objective:

*To express a security's total, systematic, and unsystematic risk in standard-deviation terms (rather than variance) for more direct interpretation.*

##### Problem:

Using Security A's total variance (39.67), beta (1.45), and the market variance (12.67), calculate the total risk, systematic risk, and unsystematic risk expressed as standard deviations.

$$\sigma_A^2 = \text{Total Risk (Variance)}$$

$$\text{Systematic Variance} = \beta_A^2 \times \sigma_m^2$$

$$\text{Unsystematic Variance} = \text{Total Variance} - \text{Systematic Variance}$$

##### Solution:

##### Step 1: Total Risk

$$\sigma_A = \sqrt{39.67} = 6.30\%$$

##### Step 2: Beta

$$\beta_A = \text{Cov}(R_A, R_m) / \sigma_m^2 = 18.33/12.67 = 1.45$$

##### Step 3: Systematic Risk

$$\text{Systematic Variance} = (1.45)^2 \times 12.67 = 26.54; \text{Systematic Risk} = \sqrt{26.54} = 5.15\%$$

##### Step 4: Unsystematic Risk

$$\text{Unsystematic Variance} = 39.67 - 26.54 = 13.13; \text{Unsystematic Risk} = \sqrt{13.13} = 3.62\%$$

##### Final Answer / Interpretation:

| Risk Component    | Variance | Standard Deviation |
|-------------------|----------|--------------------|
| Total Risk        | 39.67    | 6.30%              |
| Systematic Risk   | 26.54    | 5.15%              |
| Unsystematic Risk | 13.13    | 3.62%              |

Security A has a high beta of 1.45, meaning most of its risk comes from market movements. Of its total risk of 6.30%, 5.15% is systematic risk (market-related, cannot be diversified away) and 3.62% is unsystematic risk (firm-specific, can be diversified away).

## Section 5: Efficient Frontier Problems

### Problem 5.1: Equal Investment in Two Uncorrelated Risky Assets

#### Learning Objective:

To compute a point on the Efficient Frontier for a two-asset portfolio of risky assets whose returns are uncorrelated.

#### Problem:

An investor is considering two risky assets with a correlation of  $\rho_{12} = 0.00$ . The investor allocates 50% to Asset 1 and 50% to Asset 2. Calculate the portfolio expected return and portfolio risk.

| Asset   | Expected Return | Standard Deviation |
|---------|-----------------|--------------------|
| Asset 1 | 10%             | 7%                 |
| Asset 2 | 20%             | 10%                |

$$E(R_p) = w_1E(R_1) + w_2E(R_2)$$
$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_1\sigma_2\rho_{12}$$

#### Solution:

##### Expected Return

$$E(R_p) = (0.50)(0.10) + (0.50)(0.20) = 0.05 + 0.10 = 15\%$$

##### Portfolio Variance

$$\sigma_p^2 = (0.50)^2(0.07)^2 + (0.50)^2(0.10)^2 + 2(0.50)(0.50)(0.07)(0.10)(0)$$
$$\sigma_p^2 = 0.001225 + 0.0025 + 0 = 0.003725$$

##### Portfolio Risk

$$\sigma_p = \sqrt{0.003725} = 6.10\%$$

#### Final Answer / Interpretation:

**$E(R_p) = 15\%$ ,  $\sigma_p = 6.10\%$**

Although both assets are risky, the portfolio risk is lower because the assets are uncorrelated. This point can be plotted on the Efficient Frontier graph with risk = 6.10% on the X-axis and return = 15% on the Y-axis.

### Problem 5.2: Conservative Allocation with Low Correlation

#### Learning Objective:

*To compute a point on the Efficient Frontier for a conservative allocation between two risky assets that have a low positive correlation.*

**Problem:**

An investor wants a conservative risky-asset portfolio. The correlation between the two assets is  $\rho_{12} = 0.25$ , and the portfolio weights are  $w_1 = 70\%$ ,  $w_2 = 30\%$ . Calculate the portfolio expected return and portfolio risk.

| Asset   | Expected Return | Standard Deviation |
|---------|-----------------|--------------------|
| Asset 1 | 8%              | 6%                 |
| Asset 2 | 16%             | 12%                |

$$E(R_p) = w_1E(R_1) + w_2E(R_2)$$
$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_1\sigma_2\rho_{12}$$

**Solution:**

**Expected Return**

$$E(R_p) = (0.70)(0.08) + (0.30)(0.16) = 0.056 + 0.048 = 10.40\%$$

**Portfolio Variance**

$$\sigma_p^2 = (0.70)^2(0.06)^2 + (0.30)^2(0.12)^2 + 2(0.70)(0.30)(0.06)(0.12)(0.25)$$
$$\sigma_p^2 = 0.001764 + 0.001296 + 0.000756 = 0.003816$$

**Portfolio Risk**

$$\sigma_p = \sqrt{0.003816} = 6.18\%$$

**Final Answer / Interpretation:**

**$E(R_p) = 10.40\%$ ,  $\sigma_p = 6.18\%$**

The investor receives a higher return than with Asset 1 alone while keeping risk almost as low as Asset 1's. This shows how adding a higher-risk asset may not sharply increase portfolio risk when correlation is low.

**Problem 5.3: Aggressive Allocation with Negative Correlation**

**Learning Objective:**

*To compute a point on the Efficient Frontier for an aggressive allocation between two risky assets that are negatively correlated, demonstrating the diversification power of negative correlation.*

**Problem:**

An investor is willing to accept a higher return but wants diversification support from negatively correlated assets. The correlation is  $\rho_{12} = -0.30$ , and the portfolio weights are  $w_1 = 40\%$ ,  $w_2 = 60\%$ . Calculate the portfolio expected return and portfolio risk.

| Asset   | Expected Return | Standard Deviation |
|---------|-----------------|--------------------|
| Asset 1 | 12%             | 9%                 |
| Asset 2 | 22%             | 18%                |

$$E(R_p) = w_1E(R_1) + w_2E(R_2)$$

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\sigma_1\sigma_2\rho_{12}$$

**Solution:****Expected Return**

$$E(R_p) = (0.40)(0.12) + (0.60)(0.22) = 0.048 + 0.132 = 18.00\%$$

**Portfolio Variance**

$$\sigma_p^2 = (0.40)^2(0.09)^2 + (0.60)^2(0.18)^2 + 2(0.40)(0.60)(0.09)(0.18)(-0.30)$$

$$\sigma_p^2 = 0.001296 + 0.011664 - 0.002333 = 0.010627$$

**Portfolio Risk**

$$\sigma_p = \sqrt{0.010627} = 10.31\%$$

**Final Answer / Interpretation:**

**$E(R_p) = 18.00\%$ ,  $\sigma_p = 10.31\%$**

Although 60% of the portfolio is invested in the higher-risk asset, the negative correlation reduces total portfolio risk. This is a strong demonstration of why correlation is the real engine behind diversification.



## Section 6: Capital Market Line Problems

### Problem 6.1: Conservative Investor on the CML

#### Learning Objective:

To compute the expected return and the allocation between the risk-free asset and the market portfolio for an investor with a target portfolio risk on the Capital Market Line.

#### Problem:

| Item                             | Value |
|----------------------------------|-------|
| Risk-Free Rate $R_f$             | 4%    |
| Market Portfolio Return $E(R_M)$ | 10%   |
| Market Portfolio Risk $\sigma_M$ | 12%   |

An investor wants a portfolio risk of  $\sigma_p = 6\%$ . Calculate the expected return and the allocation between the risk-free asset and the market portfolio.

$$E(R_p) = R_f + \sigma_p [E(R_M) - R_f] / \sigma_M$$
$$w_M = \sigma_p / \sigma_M; \quad w_f = 1 - w_M$$

#### Solution:

##### CML Slope (Sharpe Ratio of the Market)

$$(10\% - 4\%) / 12\% = 0.50$$

##### Expected Return

$$E(R_p) = 4\% + 6\%(0.50) = 4\% + 3\% = 7\%$$

##### Weight in Market Portfolio

$$w_M = 6\% / 12\% = 0.50 = 50\%$$

##### Weight in Risk-Free Asset

$$w_f = 1 - 0.50 = 50\%$$

#### Final Answer / Interpretation:

Invest 50% in the risk-free asset and 50% in the market portfolio.  $E(R_p) = 7\%$ ,  $\sigma_p = 6\%$ .

This is a conservative lending portfolio because part of the wealth remains in the risk-free asset.

### Problem 6.2: Moderate Investor Holding More Market Exposure

**Learning Objective:**

*To compute the expected return and portfolio allocation on the Capital Market Line for a moderate investor with a higher target risk level than a conservative investor.*

**Problem:**

| Item                             | Value |
|----------------------------------|-------|
| Risk-Free Rate $R_f$             | 5%    |
| Market Portfolio Return $E(R_M)$ | 13%   |
| Market Portfolio Risk $\sigma_M$ | 16%   |

An investor wants a portfolio risk of  $\sigma_p = 12\%$ . Calculate the expected return and portfolio allocation.

$$E(R_p) = R_f + \sigma_p[(E(R_M) - R_f)/\sigma_M];$$

$$w_M = \sigma_p/\sigma_M; \quad w_f = 1 - w_M$$

**Solution:****CML Slope**

$$(13\% - 5\%)/16\% = 8\%/16\% = 0.50$$

**Expected Return**

$$E(R_p) = 5\% + 12\%(0.50) = 5\% + 6\% = 11\%$$

**Weight in Market Portfolio**

$$w_M = 12\%/16\% = 0.75 = 75\%$$

**Weight in Risk-Free Asset**

$$w_f = 1 - 0.75 = 25\%$$

**Final Answer / Interpretation:**

**Invest 75% in the market portfolio and 25% in the risk-free asset.  $E(R_p) = 11\%$ ,  $\sigma_p = 12\%$ .**

This is a moderate portfolio. The investor is still lending 25% at the risk-free rate but has meaningful market exposure.

**Problem 6.3: Aggressive Investor Using Leverage****Learning Objective:**

*To compute the expected return and allocation on the Capital Market Line for an aggressive investor who borrows at the risk-free rate to achieve a portfolio risk beyond that of the market portfolio.*

**Problem:**

| Item                             | Value |
|----------------------------------|-------|
| Risk-Free Rate $R_f$             | 4%    |
| Market Portfolio Return $E(R_M)$ | 12%   |
| Market Portfolio Risk $\sigma_M$ | 15%   |

An aggressive investor wants a portfolio risk of  $\sigma_p = 18\%$ . Calculate the expected return and allocation.

$$E(R_p) = R_f + \sigma_p[(E(R_M) - R_f)/\sigma_M]; \quad w_M = \sigma_p/\sigma_M; \quad w_f = 1 - w_M$$

**Solution:****CML Slope**

$$(12\% - 4\%)/15\% = 8\%/15\% = 0.5333$$

**Expected Return**

$$E(R_p) = 4\% + 18\%(0.5333) = 4\% + 9.60\% = 13.60\%$$

**Weight in Market Portfolio**

$$w_M = 18\%/15\% = 1.20 = 120\%$$

**Weight in Risk-Free Asset**

$$w_f = 1 - 1.20 = -20\%$$

**Final Answer / Interpretation:**

**Invest 120% in the market portfolio and -20% in the risk-free asset.  $E(R_p) = 13.60\%$ ,  $\sigma_p = 18\%$ .**

A negative weight in the risk-free asset means the investor borrows 20% and invests the total amount in the market portfolio. This is a leveraged portfolio, suitable only for investors with high risk tolerance and a strong ability to withstand market declines.

**Problem 6.4: Lending Portfolio between the Risk-Free Asset and the Market Portfolio****Learning Objective:**

*To compute the expected return and risk of a portfolio that lends part of an investor's wealth at the risk-free rate and invests the rest in the market portfolio.*

**Problem:**

| Item                             | Value |
|----------------------------------|-------|
| Risk-Free Rate $R_f$             | 4%    |
| Market Portfolio Return $E(R_M)$ | 9%    |

|                                  |     |
|----------------------------------|-----|
| Market Portfolio Risk $\sigma_M$ | 10% |
|----------------------------------|-----|

An investor has ₹10,00,000 and invests 40% in the risk-free asset and 60% in the market portfolio. Calculate the portfolio expected return and portfolio risk.

$$E(R_p) = w_f R_f + w_M E(R_M); \quad \sigma_p = w_M \sigma_M$$

### Solution:

#### Expected Return

$$E(R_p) = (0.40)(4\%) + (0.60)(9\%) = 1.60\% + 5.40\% = 7.00\%$$

#### Portfolio Risk

$$\sigma_p = (0.60)(10\%) = 6.00\%$$

### Final Answer / Interpretation:

**E(Rp) = 7.00%,  $\sigma_p$  = 6.00%**

The investor has reduced risk by investing part of wealth in the risk-free asset. This portfolio lies between the risk-free asset and the market portfolio on the CML — a lending portfolio suitable for a conservative investor seeking market exposure without full market risk.

## Problem 6.5: Pure Market Portfolio Position

### Learning Objective:

*To compute the expected return and risk of a portfolio positioned exactly at the market portfolio on the Capital Market Line, where the investor neither lends nor borrows.*

### Problem:

| Item                             | Value |
|----------------------------------|-------|
| Risk-Free Rate $R_f$             | 5%    |
| Market Portfolio Return $E(R_M)$ | 12%   |
| Market Portfolio Risk $\sigma_M$ | 15%   |

An investor invests 100% of wealth in the market portfolio and 0% in the risk-free asset. Calculate the portfolio expected return and portfolio risk.

$$E(R_p) = w_f R_f + w_M E(R_M); \quad \sigma_p = w_M \sigma_M$$

### Solution:

**Expected Return**

$$E(R_p) = (0)(5\%) + (1.00)(12\%) = 12\%$$

**Portfolio Risk**

$$\sigma_p = (1.00)(15\%) = 15\%$$

**Final Answer / Interpretation:**

**E(Rp) = 12%, σp = 15%**

The investor is located exactly at the market portfolio point on the CML. There is no lending and no borrowing. This is suitable for a moderate investor who is willing to accept full market risk and wants full exposure to the market portfolio.

**Problem 6.6: Borrowing Portfolio beyond the Market Portfolio****Learning Objective:**

*To compute the expected return and risk of a leveraged portfolio in which an investor borrows at the risk-free rate to invest more than 100% in the market portfolio.*

**Problem:**

| Item                             | Value |
|----------------------------------|-------|
| Risk-Free Rate $R_f$             | 4%    |
| Market Portfolio Return $E(R_M)$ | 9%    |
| Market Portfolio Risk $\sigma_M$ | 10%   |

An investor has ₹10,00,000 but borrows an additional ₹5,00,000 at the risk-free rate. Therefore, the investor invests 150% in the market portfolio and -50% in the risk-free asset. Calculate the portfolio expected return and portfolio risk.

$$E(R_p) = w_f R_f + w_M E(R_M); \quad \sigma_p = w_M \sigma_M$$

**Solution:****Expected Return**

$$E(R_p) = (-0.50)(4\%) + (1.50)(9\%) = -2.00\% + 13.50\% = 11.50\%$$

**Portfolio Risk**

$$\sigma_p = (1.50)(10\%) = 15.00\%$$

**Final Answer / Interpretation:**

$$E(R_p) = 11.50\%, \sigma_p = 15.00\%$$

The investor earns a higher expected return than the market portfolio, but risk also increases proportionately. This portfolio lies beyond the market portfolio on the CML — a leveraged borrowing portfolio suitable only for aggressive investors.

### Problem 6.7: Finding the Required Market-Portfolio Weight for a Target Risk

#### Learning Objective:

*To determine how much of a portfolio should be invested in the market portfolio, given an investor's target risk level, and to identify whether the resulting position is lending or borrowing.*

#### Problem:

| Item                             | Value |
|----------------------------------|-------|
| Risk-Free Rate $R_f$             | 4%    |
| Market Portfolio Return $E(R_M)$ | 10%   |
| Market Portfolio Risk $\sigma_M$ | 12%   |

An investor wants a portfolio risk of  $\sigma_p = 9\%$ . Find: (1) the weight in the market portfolio, (2) the weight in the risk-free asset, (3) the expected portfolio return, and (4) whether the investor is lending or borrowing.

$$w_M = \sigma_p / \sigma_M; \quad w_f = 1 - w_M; \quad E(R_p) = w_f R_f + w_M E(R_M)$$

#### Solution:

##### Weight in Market Portfolio

$$w_M = 9\% / 12\% = 0.75 = 75\%$$

##### Weight in Risk-Free Asset

$$w_f = 1 - 0.75 = 25\%$$

##### Expected Return

$$E(R_p) = (0.25)(4\%) + (0.75)(10\%) = 1.00\% + 7.50\% = 8.50\%$$

#### Final Answer / Interpretation:

$$w_M = 75\%, w_f = 25\%, E(R_p) = 8.50\%$$

The investor holds 75% in the market portfolio and 25% in the risk-free asset. Portfolio risk is lower than market risk because the investor does not fully invest in the market portfolio. This is a lending portfolio, since the investor invests part of the money in the risk-free asset.

## Problem 6.8: Kinked CML Due to a Higher Borrowing Rate

### Learning Objective:

To compute portfolio expected return under a theoretical (single rate) Capital Market Line versus a real-world kinked CML where the borrowing rate exceeds the lending rate, and to interpret why leverage becomes less attractive in practice.

### Problem:

| Item                                | Value |
|-------------------------------------|-------|
| Lending Rate / Risk-Free Rate $R_f$ | 4%    |
| Borrowing Rate $R_b$                | 6%    |
| Market Portfolio Return $E(R_M)$    | 10%   |
| Market Portfolio Risk $\sigma_M$    | 12%   |

An aggressive investor invests 130% in the market portfolio and borrows 30% at the borrowing rate. Calculate the expected return under (1) the theoretical CML assumption, where the borrowing rate = 4%, and (2) the real-world kinked-CML assumption, where the borrowing rate = 6%. Also calculate portfolio risk.

Theoretical:  $E(R_p) = w_M E(R_M) + w_f R_f$ , where  $w_M = 1.30$ ,  $w_f = -0.30$

Real-world:  $E(R_p) = w_M E(R_M) - (\text{Borrowed Weight})(R_b)$ ;  $\sigma_p = w_M \sigma_M$

### Solution:

#### Case 1: Theoretical CML

$$E(R_p) = (1.30)(10\%) + (-0.30)(4\%) = 13.00\% - 1.20\% = 11.80\%$$

#### Case 2: Real-World Kinked CML

$$E(R_p) = (1.30)(10\%) - (0.30)(6\%) = 13.00\% - 1.80\% = 11.20\%$$

#### Portfolio Risk

$$\sigma_p = (1.30)(12\%) = 15.60\%$$

### Final Answer / Interpretation:

**Theoretical CML:  $E(R_p) = 11.80\%$ . Real-world kinked CML:  $E(R_p) = 11.20\%$ .**

The higher borrowing rate reduces the return by  $11.80\% - 11.20\% = 0.60$  percentage points. This is why the CML becomes kinked beyond the market portfolio. Investors should borrow only if the additional expected return justifies the higher risk and borrowing costs — the leveraged strategy becomes less attractive when borrowing costs exceed the lending rate.





## Section 7: Security Market Line and CAPM Valuation

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### Problem 7.1: Computing a Point on the SML

**Learning Objective:**

*To compute the required return for a security using its beta, representing the exact point on the Security Market Line.*

**Problem:**

The risk-free rate is 5%. The expected market return is 9%. Security A has a beta of 1.15. Calculate the required return for Security A using CAPM.

$$E(R_i) = RFR + \beta_i[E(R_M) - RFR]$$

**Solution:**

$$\begin{aligned} E(R_i) &= 5\% + 1.15(9\% - 5\%) \\ E(R_i) &= 5\% + 1.15(4\%) = 5\% + 4.60\% = 9.60\% \end{aligned}$$

**Final Answer / Interpretation:**

**$E(R_i) = 9.60\%$**

Security A should offer a return of 9.60% to fairly compensate investors for its beta of 1.15. If Security A's expected return is exactly 9.60%, it plots on the SML and is fairly valued.

### Problem 7.2: Identifying an Undervalued Security

**Learning Objective:**

*To show that a security plotting above the SML has positive alpha and may be undervalued.*

**Problem:**

The risk-free rate is 4%. The expected market return is 12%. Security B has a beta of 1.20. An analyst estimates Security B's return at 15%. Determine whether Security B is undervalued, fairly valued, or overvalued.

$$\begin{aligned} E(R_i) &= RFR + \beta_i[E(R_M) - RFR] \\ \alpha &= \text{Estimated Return} - \text{Required Return} \end{aligned}$$

**Solution:**

**Required Return**

$$E(R_i) = 4\% + 1.20(12\% - 4\%) = 4\% + 1.20(8\%) = 4\% + 9.60\% = 13.60\%$$

**Alpha**

$$\alpha = 15\% - 13.60\% = 1.40\%$$

**Final Answer / Interpretation:**

$$\alpha = +1.40\%$$

Security B offers an estimated return of 15%, while the required return is only 13.60%. Since the estimated return exceeds the required return, Security B plots above the SML. Security B is undervalued. Decision: Buy.

**Problem 7.3: Identifying an Overvalued Security****Learning Objective:**

*To show that a security plotting below the SML has negative alpha and may be overvalued.*

**Problem:**

The risk-free rate is 6%. The expected market return is 14%. Security C has a beta of 1.50. The analyst's estimated return for Security C is 15%. Determine whether Security C is undervalued, fairly valued, or overvalued.

$$E(R_i) = RFR + \beta_i[E(R_M) - RFR]$$
$$\alpha = \text{Estimated Return} - \text{Required Return}$$

**Solution:****Required Return**

$$E(R_i) = 6\% + 1.50(14\% - 6\%) = 6\% + 1.50(8\%) = 6\% + 12\% = 18\%$$

**Alpha**

$$\alpha = 15\% - 18\% = -3\%$$

**Final Answer / Interpretation:**

$$\alpha = -3\%$$

Security C is expected to generate 15%, but investors should require 18% for its beta of 1.50. Since the estimated return is less than the required return, Security C plots below the SML. Security C is overvalued. Decision: Sell or avoid.

## Problem 7.4: Fairly Valued Defensive Security

### Learning Objective:

*To show that a low-beta (defensive) security may be correctly priced even when its expected return is lower than the market return.*

### Problem:

The risk-free rate is 5%. The expected market return is 11%. Security D has a beta of 0.60. The analyst's estimated return for Security D is 8.60%. Determine whether Security D is undervalued, fairly valued, or overvalued.

$$E(R_i) = RFR + \beta_i[E(R_M) - RFR]$$
$$\alpha = \text{Estimated Return} - \text{Required Return}$$

### Solution:

#### Required Return

$$E(R_i) = 5\% + 0.60(11\% - 5\%) = 5\% + 0.60(6\%) = 5\% + 3.60\% = 8.60\%$$

#### Alpha

$$\alpha = 8.60\% - 8.60\% = 0\%$$

### Final Answer / Interpretation:

$$\alpha = 0\%$$

Security D's estimated return exactly equals its required return. It plots directly on the SML. Although its return is lower than the market return, this is acceptable because Security D has lower systematic risk. Security D is fairly valued. Decision: Hold.

## Problem 7.5: Comparing Two Securities Using the SML

### Learning Objective:

*To show that the security with the higher expected return is not always the better investment, the better investment is the one with the higher alpha after adjusting for beta risk.*

### Problem:

The risk-free rate is 4%. The expected market return is 10%. Two securities are available. Compare the two securities using the SML and identify the better investment.

| Security   | Beta | Estimated Return |
|------------|------|------------------|
| Security E | 0.80 | 9.50%            |
| Security F | 1.40 | 12.00%           |

$$E(R_i) = RFR + \beta_i[E(R_M) - RFR]$$

$$\alpha = \text{Estimated Return} - \text{Required Return}$$

### Solution:

#### Market Risk Premium

$$E(R_M) - RFR = 10\% - 4\% = 6\%$$

#### Security E — Required Return

$$E(R_E) = 4\% + 0.80(6\%) = 4\% + 4.80\% = 8.80\%$$

#### Security E — Alpha

$$\alpha_E = 9.50\% - 8.80\% = 0.70\%$$

#### Security F — Required Return

$$E(R_F) = 4\% + 1.40(6\%) = 4\% + 8.40\% = 12.40\%$$

#### Security F — Alpha

$$\alpha_F = 12.00\% - 12.40\% = -0.40\%$$

### Final Answer / Interpretation:

**$\alpha_E = +0.70\%$ ,  $\alpha_F = -0.40\%$**

Security F has a higher estimated return, but it also carries much higher beta risk. After adjusting for systematic risk, Security E is superior because it offers a positive alpha. Decision: Buy Security E; Avoid or Sell Security F.

## Section 8: Risk-Adjusted Performance Evaluation

### Problem 8.1: Active Large-Cap Fund versus Index Fund (Endowment)

#### Learning Objective:

To compute the Sharpe ratio, Treynor's ratio, and Jensen's alpha for two competing large-cap funds and decide whether continued active management is justified on a fee-aware, risk-adjusted basis.

#### Problem:

The investment committee of a mid-sized Indian endowment is reviewing its flagship large-cap allocation. The actively managed large-cap fund returned 14% this year with annualized volatility of 19% and a beta of 1.10 against the Nifty 50. A low-cost Nifty 50 index fund returned 13% with 15.5% volatility and a beta of 1.00. The Nifty 50 itself delivered 13%, and the 91-day Treasury bill yielded 6.5%.

Compute the Sharpe ratio, Treynor's ratio, and Jensen's alpha for the active fund and the index fund, and decide whether the active mandate survives a fee-aware risk-adjusted review.

$$\text{Sharpe Ratio} = (R_p - R_f) / \sigma_p$$

$$\text{Treynor's Ratio} = (R_p - R_f) / \beta_p$$

$$\text{Jensen's Alpha} = R_p - [R_f + \beta_p(R_m - R_f)]$$

#### Solution:

##### Market Risk Premium

$$R_m - R_f = 13\% - 6.5\% = 6.5\%$$

##### Active Fund (R = 14%, $\sigma$ = 19%, $\beta$ = 1.10)

$$\text{Sharpe} = (14 - 6.5)/19 = 7.5/19 = 0.395$$

$$\text{Treynor} = (14 - 6.5)/1.10 = 7.5/1.10 = 6.82$$

$$\text{Jensen's } \alpha = 14 - [6.5 + 1.10(6.5)] = 14 - 13.65 = +0.35\%$$

##### Index Fund (R = 13%, $\sigma$ = 15.5%, $\beta$ = 1.00)

$$\text{Sharpe} = (13 - 6.5)/15.5 = 6.5/15.5 = 0.419$$

$$\text{Treynor} = (13 - 6.5)/1.00 = 6.50$$

$$\text{Jensen's } \alpha = 13 - [6.5 + 1.00(6.5)] = 13 - 13 = 0.00\%$$

| Measure        | Active Fund | Index Fund | Winner |
|----------------|-------------|------------|--------|
| Sharpe         | 0.395       | 0.419      | Index  |
| Treynor        | 6.82        | 6.50       | Active |
| Jensen's Alpha | +0.35%      | 0.00%      | Active |

**Final Answer / Interpretation:**

On systematic-risk measures (Treynor, Jensen's alpha), the active fund looks better; but on total risk (Sharpe), the index fund wins, because the active fund's higher volatility (19% vs 15.5%) pulls its Sharpe ratio below the index's. The gap between a winning Treynor and a losing Sharpe localizes the problem in unsystematic (diversifiable) risk.

Since a typical active large-cap expense ratio in India runs well above the +0.35% gross alpha, the net alpha to the endowment is almost certainly negative.

**Decision: Index the core. Redeploy to the index fund and reserve the active-risk budget for less efficient market segments where skill has more room to add value.**

**Problem 8.2: Is the PMS Manager Skilled, or Just Concentrated?****Learning Objective:**

*To compute Sharpe, Treynor, and Jensen's alpha for a concentrated portfolio and judge whether its outperformance reflects durable skill or merely higher risk-taking, given the investor's degree of diversification.*

**Problem:**

A boutique PMS manager reports a 22% return for an HNI client's concentrated equity portfolio, well above the Nifty 50's 14%. The portfolio has a beta of 1.45 and an annualized volatility of 30%, compared with the index's 16%. The 91-day Treasury bill yielded 6%. The client's PMS account represents nearly all of his equity wealth.

Compute all three measures for the PMS portfolio and the Nifty 50 benchmark, and rule on whether the 22% return is manufactured by risk or earned by skill — and whether the answer changes given this investor's undiversified position.

$$\text{Sharpe Ratio} = (R_p - R_f) / \sigma_p$$

$$\text{Treynor's Ratio} = (R_p - R_f) / \beta_p$$

$$\text{Jensen's Alpha} = R_p - [R_f + \beta_p(R_m - R_f)]$$

**Solution:****Market Risk Premium**

$$R_m - R_f = 14\% - 6\% = 8\%$$

**PMS Portfolio (R = 22%,  $\sigma$  = 30%,  $\beta$  = 1.45)**

$$\text{Sharpe} = (22 - 6) / 30 = 16 / 30 = 0.533$$

$$\text{Treynor} = (22 - 6) / 1.45 = 16 / 1.45 = 11.03$$

$$\text{Jensen's } \alpha = 22 - [6 + 1.45(8)] = 22 - 17.6 = +4.40\%$$

**Nifty 50 (R = 14%,  $\sigma$  = 16%,  $\beta$  = 1.00)**

$$\text{Sharpe} = (14 - 6)/16 = 8/16 = 0.500$$

$$\text{Treynor} = (14 - 6)/1.00 = 8.00$$

$$\text{Jensen's } \alpha = 14 - [6 + 1.00(8)] = 14 - 14 = 0.00\%$$

| Measure        | PMS    | Nifty 50 | Edge           |
|----------------|--------|----------|----------------|
| Sharpe         | 0.533  | 0.500    | +0.033 (thin)  |
| Treynor        | 11.03  | 8.00     | +3.03 (large)  |
| Jensen's Alpha | +4.40% | 0.00%    | +4.40% (large) |

#### Final Answer / Interpretation:

Treynor and Jensen's alpha both suggest strong skill, but the Sharpe ratio improves on the index by only a razor-thin 0.033. Beta-based measures ignore the diversifiable half of the portfolio's 30% volatility; Sharpe does not, and once total risk is charged for, almost all the apparent outperformance evaporates.

Because this HNI holds the PMS as nearly all of his equity wealth, there is no "rest of the portfolio" to diversify against — his relevant scoreboard is Sharpe, not Treynor or Jensen's alpha.

**Decision: Do not raise the commitment; if anything, cap it, and recommend trimming the PMS to a sleeve while diversifying the remainder.**

### Problem 8.3: Should the Mid-Cap Fund Be Included in SIP Recommendations?

#### Learning Objective:

*To compute Sharpe, Treynor, and Jensen's alpha for a mid-cap fund against its correct benchmark, and to assess suitability for disciplined, long-horizon SIP investors in light of risks the ratios do not capture.*

#### Problem:

A wealth advisory firm is deciding whether to push a popular mid-cap fund into its monthly SIP recommendations. The fund returned 19% with annualized volatility of 30% and a beta of 1.20 against the Nifty Midcap 150, which itself returned 16% with 22% volatility. The 91-day Treasury bill yielded 6%.

Compute all three measures for the fund relative to its correct mid-cap benchmark, then assess SIP suitability in light of volatility, drawdown, and beta instability.

$$\text{Sharpe Ratio} = (R_p - R_f) / \sigma_p$$

$$\text{Treynor's Ratio} = (R_p - R_f) / \beta_p$$

$$\text{Jensen's Alpha} = R_p - [R_f + \beta_p(R_m - R_f)]$$

### Solution:

#### Market (Benchmark) Risk Premium

$$R_m - R_f = 16\% - 6\% = 10\%$$

#### Mid-Cap Fund (R = 19%, $\sigma$ = 30%, $\beta$ = 1.20)

$$\text{Sharpe} = (19 - 6)/30 = 13/30 = 0.433$$

$$\text{Treynor} = (19 - 6)/1.20 = 13/1.20 = 10.83$$

$$\text{Jensen's } \alpha = 19 - [6 + 1.20(10)] = 19 - 18 = +1.00\%$$

#### Nifty Midcap 150 (R = 16%, $\sigma$ = 22%, $\beta$ = 1.00)

$$\text{Sharpe} = (16 - 6)/22 = 10/22 = 0.455$$

$$\text{Treynor} = (16 - 6)/1.00 = 10.00$$

$$\text{Jensen's } \alpha = 16 - [6 + 1.00(10)] = 16 - 16 = 0.00\%$$

| Measure        | Mid-Cap Fund | Midcap 150 | Winner    |
|----------------|--------------|------------|-----------|
| Sharpe         | 0.433        | 0.455      | Benchmark |
| Treynor        | 10.83        | 10.00      | Fund      |
| Jensen's Alpha | +1.00%       | 0.00%      | Fund      |

#### Final Answer / Interpretation:

Against its own benchmark, the fund shows a positive alpha and a superior Treynor, but a Sharpe ratio that falls short of the plain index — the excess volatility (30% vs 22%) is the drawdown risk that compliance is worried about. All three measures assume a stable beta, but mid-cap betas drift across cycles, so the 10.83 Treynor and +1.00% alpha inherit that fragility.

**Decision: Recommend the fund for SIPs only with guardrails — as a modestly sized satellite for long-horizon, disciplined investors — not as a default recommendation. The sub-benchmark Sharpe and unstable beta are disqualifying for risk-averse or short-horizon clients.**

### Problem 8.4: Two Pension Managers, Three Conflicting Signals

#### Learning Objective:

*To compute Sharpe, Treynor, and Jensen's alpha for two managers with identical returns but different risk profiles, resolve a ranking reversal, and identify which measure is decision-relevant for a diversified institutional investor.*

#### Problem:

A pension fund's investment committee is reviewing two external institutional equity managers for renewal; the budget allows retaining only one. Manager X returned 15% with an annualized volatility of



14% and a beta of 1.00 against the Nifty 50. Manager Y also returned 15%, but with 18% volatility and a beta of 0.85. The Nifty 50 returned 13%, and the 91-day Treasury bill yielded 6%.

Compute all three measures for each manager, identify the ranking reversal, and decide which manager a diversified pension pool should retain.

$$\begin{aligned}\text{Sharpe Ratio} &= (R_p - R_f) / \sigma_p \\ \text{Treynor's Ratio} &= (R_p - R_f) / \beta_p \\ \text{Jensen's Alpha} &= R_p - [R_f + \beta_p(R_m - R_f)]\end{aligned}$$

### Solution:

#### Market Risk Premium

$$R_m - R_f = 13\% - 6\% = 7\%$$

#### Manager X (R = 15%, $\sigma$ = 14%, $\beta$ = 1.00)

$$\begin{aligned}\text{Sharpe} &= (15 - 6)/14 = 9/14 = 0.643 \\ \text{Treynor} &= (15 - 6)/1.00 = 9.00 \\ \text{Jensen's } \alpha &= 15 - [6 + 1.00(7)] = 15 - 13 = +2.00\%\end{aligned}$$

#### Manager Y (R = 15%, $\sigma$ = 18%, $\beta$ = 0.85)

$$\begin{aligned}\text{Sharpe} &= (15 - 6)/18 = 9/18 = 0.500 \\ \text{Treynor} &= (15 - 6)/0.85 = 9/0.85 = 10.59 \\ \text{Jensen's } \alpha &= 15 - [6 + 0.85(7)] = 15 - 11.95 = +3.05\%\end{aligned}$$

| Measure        | Manager X | Manager Y | Winner |
|----------------|-----------|-----------|--------|
| Sharpe         | 0.643     | 0.500     | X      |
| Treynor        | 9.00      | 10.59     | Y      |
| Jensen's Alpha | +2.00%    | +3.05%    | Y      |

### Final Answer / Interpretation:

Same return (15%), opposite verdicts. Manager X is the efficient diversifier — low total volatility, market-like beta, superior Sharpe. Manager Y delivered the identical return while taking less market risk (beta 0.85); most of Y's higher total volatility is idiosyncratic, which is exactly what sinks Y's Sharpe. Because a pension fund is a large, diversified pool, a single manager's idiosyncratic risk diversifies away at the fund level, so Treynor and Jensen's alpha are the decision-relevant measures here.

**Decision: Retain Manager Y, conditional on verifying that Y's idiosyncratic bets are genuinely uncorrelated with the rest of the manager roster. (Had this been a single-manager, undiversified mandate, Manager X — with the superior Sharpe — would have been the correct choice instead.)**

## Problem 8.5: Family Office: Where Does the Next Rupee Go?

**Learning Objective:**

To compute Sharpe, Treynor, and Jensen's alpha for three portfolio sleeves, and to allocate capital by weighing the ratios against governance constraints such as benchmark validity, liquidity, fees, and suitability.

**Problem:**

A family office is allocating fresh capital across three existing equity sleeves and must decide which deserves a top-up. A Category III AIF returned 20% with an annualized volatility of 24% and a beta of 1.30 against the Nifty 50. A Nifty 50 ETF returned 13% with 15% volatility and a beta of 1.00. A conservative balanced equity portfolio returned 12% with 11% volatility and a beta of 0.65. The Nifty 50 returned 13%, and the 91-day Treasury bill yielded 6%.

Compute all three measures for each sleeve, then decide which merits the additional allocation after weighing governance constraints against the ratios.

$$\text{Sharpe Ratio} = (R_p - R_f) / \sigma_p$$

$$\text{Treynor's Ratio} = (R_p - R_f) / \beta_p$$

$$\text{Jensen's Alpha} = R_p - [R_f + \beta_p(R_m - R_f)]$$

**Solution:****Market Risk Premium**

$$R_m - R_f = 13\% - 6\% = 7\%$$

**Category III AIF (R = 20%,  $\sigma$  = 24%,  $\beta$  = 1.30)**

$$\text{Sharpe} = (20 - 6)/24 = 14/24 = 0.583$$

$$\text{Treynor} = (20 - 6)/1.30 = 14/1.30 = 10.77$$

$$\text{Jensen's } \alpha = 20 - [6 + 1.30(7)] = 20 - 15.1 = +4.90\%$$

**Nifty 50 ETF (R = 13%,  $\sigma$  = 15%,  $\beta$  = 1.00)**

$$\text{Sharpe} = (13 - 6)/15 = 7/15 = 0.467$$

$$\text{Treynor} = (13 - 6)/1.00 = 7.00$$

$$\text{Jensen's } \alpha = 13 - [6 + 1.00(7)] = 13 - 13 = 0.00\%$$

**Balanced Equity Portfolio (R = 12%,  $\sigma$  = 11%,  $\beta$  = 0.65)**

$$\text{Sharpe} = (12 - 6)/11 = 6/11 = 0.545$$

$$\text{Treynor} = (12 - 6)/0.65 = 6/0.65 = 9.23$$

$$\text{Jensen's } \alpha = 12 - [6 + 0.65(7)] = 12 - 10.55 = +1.45\%$$

| Measure        | AIF    | ETF   | Balanced | Rank                 |
|----------------|--------|-------|----------|----------------------|
| Sharpe         | 0.583  | 0.467 | 0.545    | AIF > Balanced > ETF |
| Treynor        | 10.77  | 7.00  | 9.23     | AIF > Balanced > ETF |
| Jensen's Alpha | +4.90% | 0.00% | +1.45%   | AIF > Balanced > ETF |

**Final Answer / Interpretation:**

On every ratio the AIF ranks first and the ETF last — but this clean sweep is misleading. A Category III AIF's risk is poorly described by the Nifty 50, so its beta is mismeasured and its Treynor and Jensen's alpha figures are unreliable. Only the Sharpe ratio, which needs no benchmark, is cleanly comparable, and there the AIF's edge over the balanced sleeve is real but far narrower (0.583 vs 0.545).

The ETF's near-zero alpha is a feature, not a failure — by construction it delivers market beta. The balanced sleeve posts the second-best Sharpe with the lowest volatility and beta, the profile most aligned with capital preservation. AIFs also carry lock-ins, illiquidity, and high fees that the gross returns above do not net out.

**Decision: Split the top-up. Direct the core of the new capital to the balanced sleeve, keep the ETF as the liquid, low-cost beta anchor, and allow only a capped, sleeve-sized increment to the AIF, contingent on a proper strategy benchmark, fee transparency, and acceptable liquidity terms.**